

What a rainfall flood model can and cannot be trusted to decide

A blind reconstruction of the Franklin and Licking County flood, tested against what actually happened — and a specific, quantified limit on using regional flood screening for road-closure decisions.

This revision adds the completed six- and eight-metre corridor simulations, which both confirm the central finding and place a limit on it.

Prepared 23 August 2026 · **revised** 26 August **Domain** Greater Columbus & western Licking County

Corridor runs 2 of 3 complete at 6–8 m **Forcing** NOAA MRMS radar QPE **Terrain** USGS 3DEP 1 m LiDAR

Status Research — not operationally validated

What this document is. An independent technical study produced with a research flood simulator. It is not a product of, endorsed by, or reviewed by ODOT, the National Weather Service, USGS, or any Ohio agency. Observed impacts are drawn from public reporting and public data; several timestamps are approximate or inferred and are labelled as such. Nothing here should be used as the basis of an operational decision without agency review and independent verification.

BOTTOM LINE

HOLDS UP

Regional screening works. Driven only by published rainfall and public terrain — with predictions frozen before any flood report was consulted — the model placed 16 of 17 documented flood locations among its deepest water. Against a matched random baseline that outcome would occur by chance about 3–7% of the time.

DOES NOT HOLD UP

Road-level closure calls from a 60 m grid are unsafe. At the I-70 crossing of the South Fork Licking River, the coarse model declares the interstate impassable while its own water surface sits **1.77 m below the actual pavement**. The flooding it sees is the channel, not the roadway.

RETRACTED

The apparent 11.6-hour warning on the I-70 closure does not survive scrutiny. The model's early signal arrived before the upstream river gauge had even peaked, so it cannot be the flood wave that closed the road. We report it as a false positive, not a lead time.

NEWLY QUALIFIED

An elevation mismatch marks a closure as unverified, not as false. Two corridors were re-run at six and eight metres. Where the flooding would have to come from a channel, the coarse signal was smearing — at SR-310 the stream carried 4.28 m while the road stayed below its closure threshold. But at W Broad Street, which has the largest elevation mismatch in the study, the road floods anyway, because rain ponds there and no stream is present. Five of twelve corridor points could be scored: four confirmed, one refuted.

OPERATIONALLY USEFUL

The observed routing lag is the real warning asset. The South Fork crest took roughly 28 hours to travel from Kirkersville to the Buckeye Lake reach. That lag — measured, not modelled — is genuine, actionable lead time for the I-70 corridor.

01 What was tested, and how the test was kept honest

Cause data in, predictions frozen, impacts gathered only afterwards

The simulator was given two things and nothing else: the rainfall that fell, and the shape of the ground. Rainfall came from NOAA's Multi-Radar Multi-Sensor system — both the low-latency radar-only product that would have been available in real time, and the gauge-corrected retrospective product. Terrain came from USGS 3DEP, at 1 m resolution over the study corridors.

The order of operations is what makes the result meaningful. Sensor locations were placed automatically by the terrain itself, not by hand. The predictions were then sealed under content-derived identifiers. Only after that were the seventeen documented flood locations collected from public reporting and scored by a script whose thresholds were fixed in

advance. No rainfall, roughness, terrain, or threshold was ever adjusted to improve agreement with a known outcome.

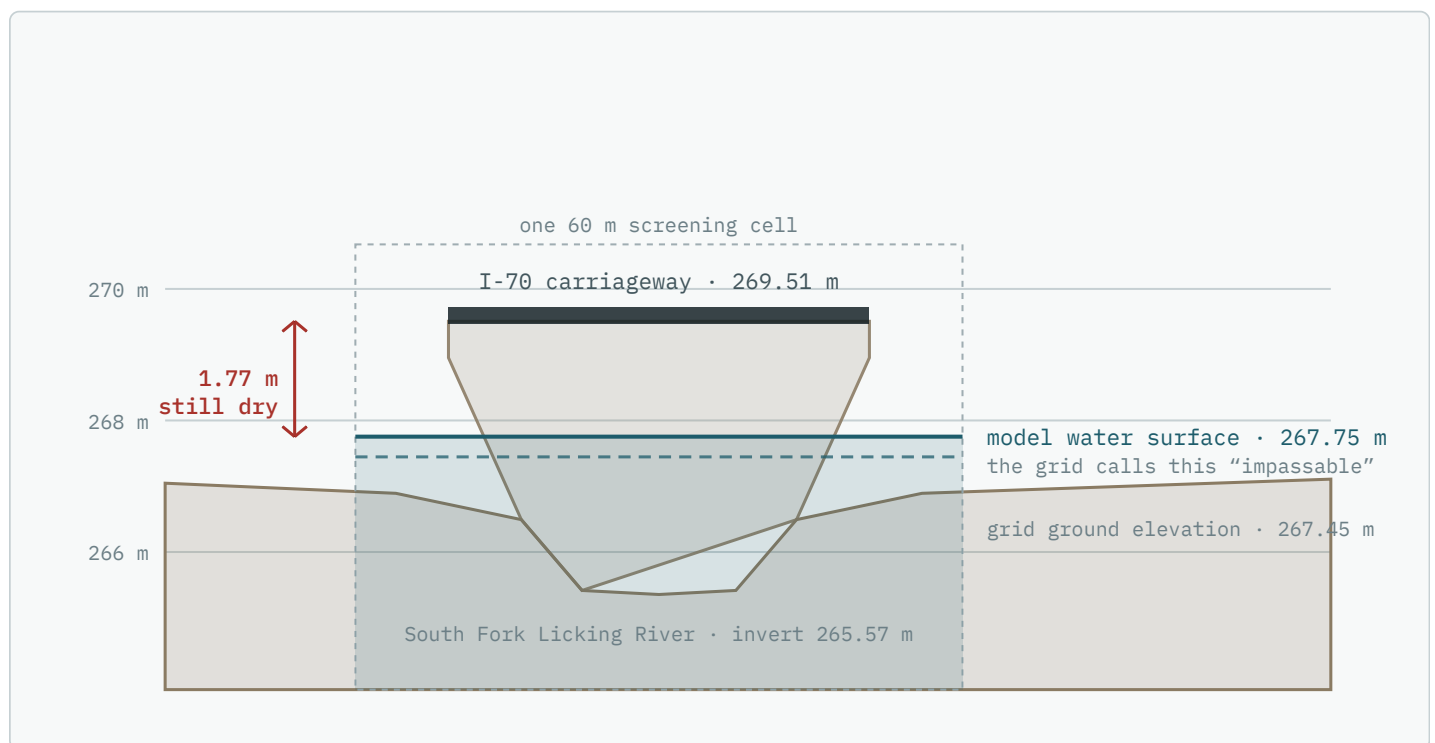
Rainfall was checked against gauges over **matched accumulation windows** and the difference reported rather than corrected. Over the storm window the retrospective product landed within 4% of the Buckeye Lake observation and inside the spread of the Pataskala gauges; the radar-only product ran materially low in this warm-rain convective event, which is a real limitation of that feed and is carried forward as such.

02 The finding that matters for infrastructure

A 60-metre cell cannot tell a roadway from the stream beneath it

A regional screening grid divides the landscape into 60-metre squares and gives each square a single ground elevation. Where a highway crosses a stream, that one square contains the carriageway, the embankment, and the channel bed all at once. Whichever elevation the grid adopts becomes the level at which "the road" floods in the model.

At the I-70 crossing of the South Fork Licking River, the grid adopted an elevation of 267.45 m — roughly halfway between the channel invert at 265.57 m and the true carriageway at 269.51 m. The consequence is direct and measurable.



Surveyed section versus screening grid, I-70 at the South Fork Licking River. Ground profile and carriageway elevation from USGS 3DEP 1 m LiDAR; grid elevation and water surface from the 60 m screening run at the moment it crossed its impassability threshold. The model is not wrong about there being water — it is wrong about where that water is.

This is not a single unlucky location. Applying the same measurement across twelve monitored corridor points shows the error runs in both directions, which matters: a coarse grid can invent a closure, and it can also miss one.

CORRIDOR POINT	GRID WATER SURFACE AT "IMPASSABLE"	SURVEYED PAVEMENT	FREEBOARD	READING
W Broad St, Hilltop	218.82	222.97	+4.15	tested at 6 m – road really floods
SR-310 at South Fork	313.30	316.73	+3.44	floods a dry road
I-70 at South Fork	267.75	269.51	+1.77	floods a dry road
SR-79 / Hebron Rd	269.49	269.41	-0.08	grid ≈ pavement
Main St at Broad, Pataskala	303.04	302.86	-0.18	grid ≈ pavement
US-40, Hebron	271.57	271.28	-0.29	grid ≈ pavement
South Fork at Kirkersville	280.01	279.65	-0.36	grid ≈ pavement
Franklinton core	215.78	215.40	-0.38	grid ≈ pavement
US-40, east of Etna	297.44	296.89	-0.55	under-reports
I-70 at SR-79	269.86	269.16	-0.71	under-reports
US-40, Etna	325.04	321.86	-3.18	under-reports
Olentangy at Columbus	221.95	216.91	-5.04	under-reports

All elevations in metres, NAVD88. **Freeboard** is the height of the surveyed carriageway above the water surface at which the 60 m grid declares that road impassable. A positive value means the grid reports a flooded road while the pavement is still dry by that margin. A negative value means the grid's ground sits above the real road, so genuine street flooding goes unseen. On elevation alone, three of twelve points look like invented closures and four look like missed ones. **Two of those twelve have since been tested against fine-resolution runs, and one of the three suspected invented closures did not survive the test** — see the next section.

03 What the fine-resolution runs actually settled

The elevation test finds where a coarse grid *can* be wrong, not where it *is*

The freeboard measurement above is a statement about terrain. It says nothing about whether water ever arrives. To find out, two corridors were re-run at six and eight metres — fine enough to separate a carriageway from the channel beside it — each carrying radar rainfall on every cell and taking its boundary water levels from the coarse model around it.

Both runs completed their full twenty-hour window. They did not simply confirm the elevation test. One of them broke it, and that is the more useful result.

CORRIDOR POINT	FREEBOARD	WHAT ELEVATION ALONE PREDICTED	FINE-GRID ROAD DEPTH	CHANNEL BESIDE IT	OUTCOME
SR-310 at South Fork	+3.44	floods a dry road	0.20	4.28	confirmed
Main St at Broad, Pataskala	-0.18	grid $\approx$ pavement	3.69	4.47	confirmed
South Fork at Kirkersville	-0.36	grid $\approx$ pavement	2.99	2.99	confirmed
Franklinton core	-0.38	grid $\approx$ pavement	1.70	—	confirmed
W Broad St, Hilltop	+4.15	floods a dry road	2.22	—	prediction failed

Depths in metres. Only these five of the twelve corridor points can be scored: four lie in the Buckeye Lake corridor, whose fine run is held for off-platform completion; one sits on a model boundary where its water level is imposed rather than computed; and two carry a prediction that the coarse grid *misses* flooding, which cannot be confirmed because the fine grid does not flood those roads either. **Five scoreable, four confirmed, one failed.**

SR-310 is the mechanism caught in the act. The coarse grid calls that crossing impassable. At eight metres the channel beside it carries 4.28 m of water while the carriageway never reaches the 0.30 m closure threshold — a difference of over four metres between the road and the stream the coarse cell had averaged together. That is exactly the error the elevation test was designed to detect, and here it is, operating.

W Broad Street is the correction. It carries the largest freeboard measured anywhere in this study: its coarse cell sits 4.15 m below the pavement, so on elevation alone it looked like the clearest invented closure of the twelve. At six metres the road floods regardless, to 2.22 m. The terrain explains why — that location has *no stream in it at all*. The water is rainfall ponding in a local low point on a hilltop, not channel water borrowed from a neighbouring cell.

A large elevation mismatch marks a coarse-grid closure as *unverified*, not as false. Where the flooding would have to come from a channel, the signal is likely smearing. Where rain can pond on the spot, it may be real.

The distinguishing question — is there a watercourse in this cell — is answerable from maps, before any model is run

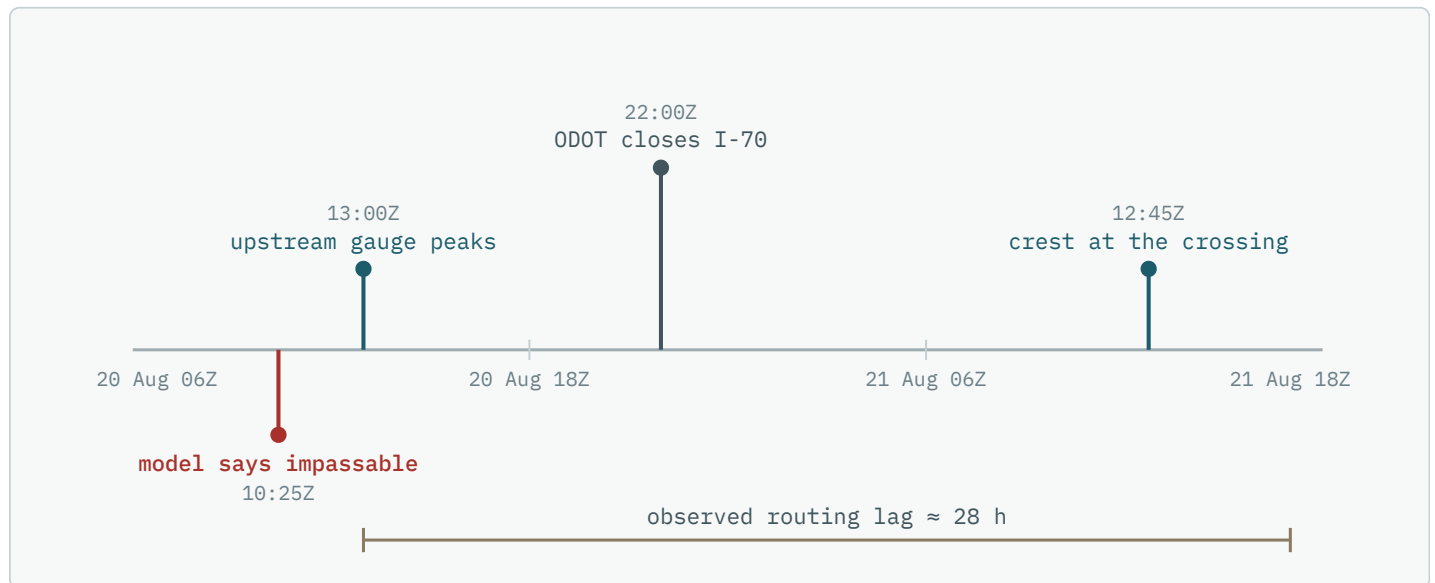
One further number deserves to be stated with its own caveat rather than quoted. In the Franklinton run the modelled river peak at the Olentangy gauge lands within **31 minutes** of the observed peak, with a shape correlation of 0.917. That is a good number and it is close to meaningless as a test: that gauge sits on the edge of the fine model's domain, where its water level is dictated by the coarse model rather than computed. The honest comparison is the Pataskala gauge, which sits well inside its domain and peaks **153 minutes early**. The fine grid resolves *where* water goes considerably better than *when* it arrives.

04 Why the early I-70 signal was not a warning

A flood wave cannot arrive downstream before it peaks upstream

The screening run crossed its impassability threshold at the I-70 corridor at 10:25Z on 20 August, roughly eleven and a half hours before ODOT closed the interstate. Taken at face value that looks like a substantial early warning. It is not, and the reason is visible in the observed river record rather than in the model.

The South Fork Licking River gauge upstream at Kirkersville did not peak until 13:00Z — two and a half hours *after* the model's signal. The downstream response in the Buckeye Lake reach did not peak until the following afternoon, a routing lag of about twenty-eight hours. Water that closed I-70 had not yet been delivered to the upstream gauge when the model announced the road was impassable.



Sequence of the South Fork Licking response. River timings from NOAA National Water Model analysis fields assimilated to live USGS gauges; closure time as announced by ODOT and reported in press coverage; crest elevation and timing from National Weather Service product text quoted in reporting. The model's signal precedes the upstream peak, which rules it out as the same physical event.

We are withdrawing the eleven-and-a-half-hour lead time. The model was not early about the closure; it was reporting a different kind of water, in a place its grid could not resolve.

Formal retraction — recorded in the study's determination file

05 What the model could not do at all

Multi-day river routing is outside its represented physics

Extending the simulation to fifty-six hours to reach for the observed crest exposed a structural defect rather than a tuning problem. After the rainfall stops, water at the I-70 corridor recedes at a constant rate that exactly matches the model's drainage parameter, instead of rising toward the crest that actually arrived a day later.

Two causes, both identified before any comparison to the observed crest. The model's storm-drain term removes water uniformly across the domain, so it drains routed river water that should be travelling downstream. And nothing in the configuration represents lake storage, dam or spillway operation, baseflow, or inflow entering from beyond the domain boundary. The twenty-eight-hour lag through the Buckeye Lake system is precisely the process the configuration omits.

This is reported as a discovered defect. A sensitivity run isolating the drainage term is queued, and it will quantify the term's contribution rather than fix the outcome.

06 What this means for Ohio infrastructure decisions

Four recommendations that follow directly from the measurements above

RECOMMENDATION 01

Use regional flood screening for staging, not for closures

The 60-metre screen reliably identifies which corridors and neighbourhoods will be in trouble, which is exactly what resource pre-positioning, crew staging, and shelter activation need. It should not be permitted to generate road-open or road-closed guidance, in either direction, at any crossing.

RECOMMENDATION 02

Tie closure thresholds to surveyed pavement elevations

Every point where the model's ground elevation and the surveyed carriageway diverge by more than half a metre is a place where automated guidance will be wrong. The fix is inexpensive: pair each flood-prone crossing with its surveyed pavement elevation and evaluate the modelled water surface against that number, not against a grid cell.

RECOMMENDATION 03

Treat upstream gauges as the warning instrument for the I-70 corridor

The measured twenty-eight-hour travel time between the Kirkersville reach and the Buckeye Lake reach is real lead time available today, from instruments already in the ground. For this corridor a gauge-threshold rule offers warning that no rainfall model in this study matched.

RECOMMENDATION 04

Verify culvert and bridge openings before trusting fine-scale results

Detecting stream crossings automatically from elevation data reached 91–96% precision against 1 m LiDAR across three corridors, which is useful but not sufficient. A crossing wrongly modelled as solid embankment creates a dam that does not exist — and a high-resolution model can be confidently wrong in a way a coarse one is not.

07 Limitations

Stated plainly, because they bound every claim above

- **No false-alarm rate.** Public reporting records what flooded, never what stayed dry. Without an authoritative record of roads that remained open, the model's tendency to over-predict cannot be quantified — only its detection can.
- **River observations are a proxy.** Direct USGS endpoints were unreachable from the study environment. River timings come from National Water Model analysis fields that assimilate live USGS gauges — closely anchored to observation, but not raw observation, and carrying discharge only. No stage comparison, and therefore no stage error statistic, is possible here.
- **Several impact timestamps are approximate.** Closure and rescue times come from press accounts, some phrased as "around" a given hour, and at least one closure start is inferred rather than stated. Two independently reported accounts of the I-70 closure time disagree by roughly ninety minutes.
- **Only five of twelve corridor points can be scored.** Two of the three fine-scale corridor models are complete. The third, covering the Buckeye Lake and I-70 crossings, is held mid-run for completion on other hardware, which leaves four points untested — including the I-70 crossing that opens this document. One further point sits on a model boundary where its water level is imposed rather than computed, and two carry a prediction that the coarse grid *misses* flooding, which cannot be confirmed because the fine grid does not flood those roads either.
- **The fine grid's timing is worse than its geography.** At the one gauge that sits well inside a fine domain, the modelled peak arrives **153 minutes early**. The fine models resolve where water goes considerably better than when it arrives, and nothing here should be read as a timing capability.

- **One fine-scale run passed through a numerically strained phase.** During the Franklinton rainfall peak the solver briefly carried flow at Froude numbers around 3.5, outside the validity range of the simplified momentum equation it uses. Velocities in that window are not defensible; depths, which the closure thresholds actually use, are far less sensitive to it, and the condition relaxed as the rivers filled. It is recorded rather than smoothed over.
- **Buildings are not represented.** Building footprint sources were unreachable from the study environment, so urban flow obstruction in Franklinton is modelled by terrain alone.
- **One methodological change followed exposure to outcome data.** The simulation window was extended after the observed crest timing was known. It is logged as such in the study record. All forcing, geometry, roughness, and thresholds were fixed beforehand and never altered.

Data sources. Rainfall: NOAA Multi-Radar Multi-Sensor QPE, radar-only 15-minute and multi-sensor hourly products, retrieved from the NOAA Open Data bucket with per-file checksums. Terrain: USGS 3DEP 1 m and 1/3-arcsecond elevation. Hydrography: USGS National Hydrography Dataset. Roads: USGS National Transportation Dataset. River response: NOAA National Water Model version 3 analysis fields. Impacts, closures, and warning times: public reporting and National Weather Service product text quoted therein.

Predictions were sealed under content-derived run identifiers before outcome data was collected, and the analysis is reproducible from the recorded inputs. Prepared as independent research; not an official product of any agency.